

# Dewell Guerand

Belgium | [guerand.com](http://guerand.com) | [github.com/DewellGuerand](https://github.com/DewellGuerand) | [linkedin.com/GuerandDewell](https://linkedin.com/GuerandDewell)

## Education

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|---------------------------------------------------------------------------|-----------------------|
| <b>MEng Double Degree</b> , CentraleSupélec, Université Paris-Saclay      | Aug 2026 – June 2028  |
| • Engineering cycle – T.I.M.E (Top International Managers in Engineering) |                       |
| <b>MEng</b> , Louvain School of Engineering (EPL), UCLouvain              | Sept 2025 – June 2026 |
| • Major: Applied Mathematics                                              |                       |
| <b>BEng</b> , Louvain School of Engineering (EPL), UCLouvain              | Sept 2022 – June 2025 |
| • Major: Applied Mathematics, Minor: Computer Science                     |                       |
| • Graduated <i>Cum Laude</i>                                              |                       |
| <b>Lycée de Berlaymont</b> , Belgium                                      | Sept 2016 – June 2022 |
| • Graduated with honors – Major in Mathematics and Sciences               |                       |

## Experience

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| <b>Quant Analyst</b> , LSM Investment Club – UCLouvain                                                                                       | Oct 2025 – Present |
| • Conducted quantitative analysis and financial modelling to support investment decisions across equity and fixed income strategies.         |                    |
| • Presented and popularized recent academic research in quantitative finance to club members; contributed to portfolio strategy discussions. |                    |

## Projects

|                                                                                                                                                                                                                                                                  |                                                                         |
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| <b>Interest Rate Modeling – Sinusoidal Hull-White Model</b>                                                                                                                                                                                                      | <a href="https://github.com/DewellGuerand">github.com/DewellGuerand</a> |
| • Implemented and validated an interest rate simulation engine based on the Hull-White model, incorporating a sinusoidal mean-reversion extension derived from a research paper; calibrated model parameters and benchmarked against standard short-rate models. |                                                                         |
| • Tools: Python (NumPy, SciPy, Matplotlib)                                                                                                                                                                                                                       |                                                                         |

### Pricer for GMAB & GMDb Insurance Products

- Developed a full pricing framework for Guaranteed Minimum Accumulation Benefit (GMAB) and Guaranteed Minimum Death Benefit (GMDb) contracts, covering closed-form analytical derivations and numerical methods including Binomial Trees and Monte Carlo simulation (Euler-Maruyama scheme); compared method accuracy and computational efficiency.
- Tools: Python (NumPy, SciPy)

### Disease Detection – Machine Learning

- Designed and trained a disease classification pipeline combining a CNN for image-based features with ensemble methods (XGBoost, RandomForest); performed hyperparameter tuning and cross-validation. Tools: Python (TensorFlow, scikit-learn)

### Finite Element Analysis – 2D Linear Elasticity

- Developed a complete finite element analysis application for solving 2D linear elasticity problems in engineering structures, with mesh generation, solver, and visualization components.
- Tools: C, Python

## Personal Achievements

- **Hockey:** 15+ years of competitive play at club level; coached and mentored junior players.
- **Running:** Completed the Brussels 20 km (half-marathon distance), training consistently while managing a full academic workload.
- **Market Investment:** Actively managing a personal portfolio; developed interest in equity markets, factor investing, and macroeconomic analysis.
- **Senior Prom Trek:** Ranked 13<sup>th</sup> out of 120 High schools in the Walloon region in a 40 km endurance trek.

## Skills & Languages

**Programming:** Python, C, C++, Java, Dart, Bash, Oz, Julia  
**Libraries:** NumPy, SciPy, Matplotlib, Pandas, TensorFlow, scikit-learn  
**Tools:** Git, Overleaf, Typst  
**Languages:** French (C2, native) | English (C1, IELTS 7.0) | Dutch (B2)